

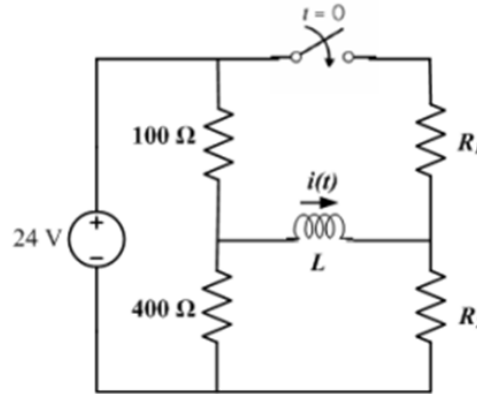
EEC-201 NETWORK THEORY

Time: 3 Hrs.

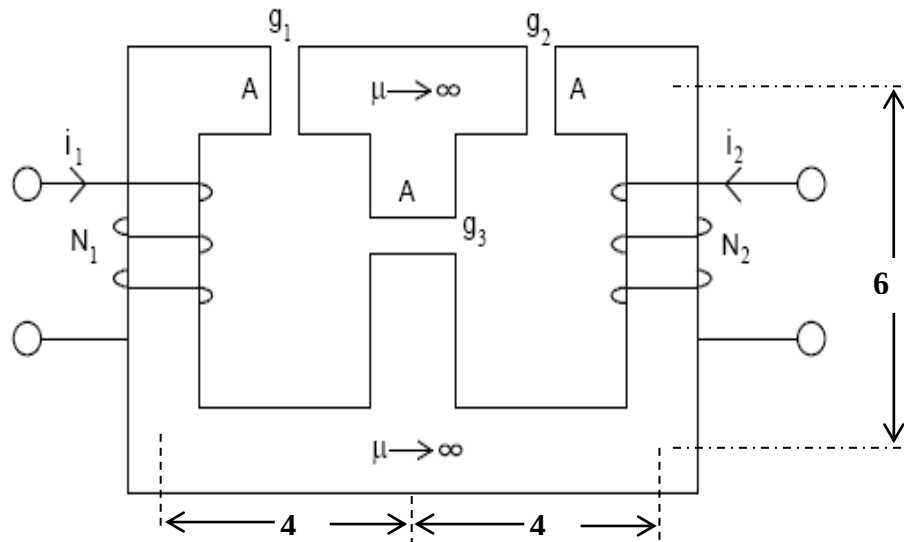
Max. Marks: 90

Answer all questions.

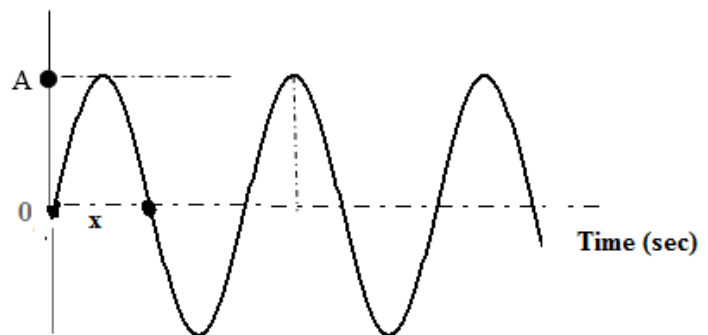
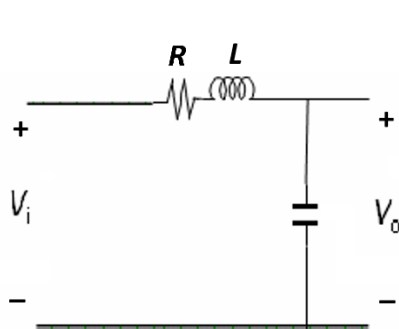
- 1) For the circuit shown, before $t = 0$ the switch was in open position for a long time. At $t = 0$, the switch is closed. The current $i(t)$ is given by $i(t) = 15 + 53.6e^{-548t} \text{ mA}$ for $t > 0$. Determine the values of R_1 , R_2 and L . [12]



- 2) Figure shows a magnetic circuit with air gaps $g_1 = g_2 = g_3 = 1 \text{ mm}$ and coils $N_1 = 200$ turns and $N_2 = 100$ turns. The cross sectional area A of the circuit is $20\text{mm} \times 10\text{mm}$. The currents in the coils are $i_1 = i_2 = i$. Assume the permeability of the core material is infinity. The fringing effect is **not** neglected. Determine the current i if the flux in the air gap g_3 is $8 \times 10^{-5} \text{ wb}$ downwards. [10]



- 3) Using the network function approach, find the sinusoidal steady-state output $V_o(t)$ of the circuit shown in Fig.a. The input $V_i(t)$ is shown in Fig. b where $A = 2 \text{ V}$ and $x = 1.0471 \text{ sec}$. $R = 2 \Omega$, $L = 1 \text{ H}$, $C = 0.5 \text{ F}$ [10]



(a)

(b)

- 4) A MATLAB code is given below.

```
s=tf([9],[1 3 9]);
```

```
step(s,10)
```

(i) Roughly sketch the output of the code.

(ii) What is the time constant?

(iii) Assuming that the response reaches the steady value after 6 time constants, what is the steady state value of the MATLAB output?

(iv) Without analytically obtaining the output, how can you predict the nature of the response? [10]

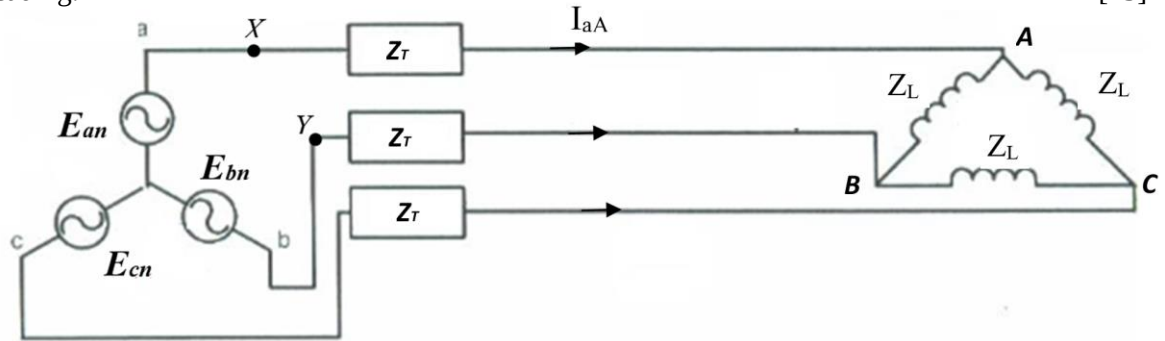
- 5) Consider the system $\dot{x} = Ax$, $y = Cx$. Suppose that the matrix A is partially specified as $A = \begin{bmatrix} \alpha & -2 \\ 1 & 0 \end{bmatrix}$. Find

the matrix A and the initial state $x(0)$ which gives an output response $y(t) = 4e^{-t}$ when $C = \begin{bmatrix} -1 & 1 \end{bmatrix}$. Also, find

the output matrix C for which the output response is $y(t) = 4e^{-t}$ when $x(0) = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$. [12]

- 6) The figure shows a 3-phase balanced ABC system of line to neutral voltage $E_{an} = 250$ V supplies a balanced delta-connected load Z_L through a transmission line with impedance Z_T . If $Z_L = 60 - j60 \Omega$ and $Z_T = 5 + j5 \Omega$, determine the line current I_{aA} , load current I_{BC} and the line voltage E_{AB} . Take E_{an} as reference.

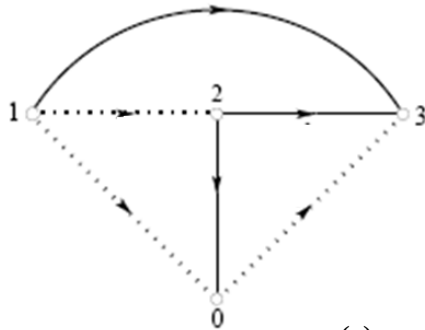
A wattmeter is connected with current coil in line a and pressure coil between the points X and Y . Determine the wattmeter reading. [15]



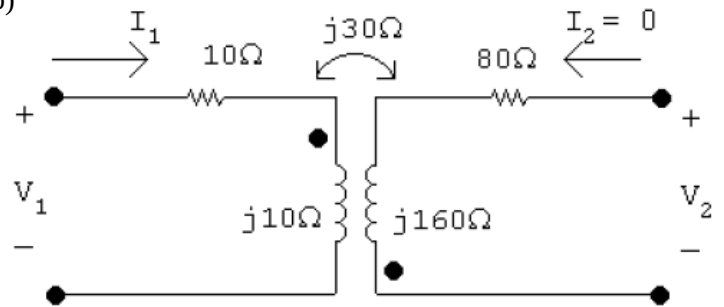
- 7) (a) If a balanced voltage $v(t) = 1.2 \cos \omega t$ V supplies a nonlinear load. If the load current $i(t)$ is $0.6 \cos(\omega t + 30^\circ) + 0.1 \cos(5\omega t + 45^\circ) + 0.1 \cos(7\omega t + 60^\circ)$ A, determine average power, %THD and true power factor.

(b) Show that all the branch voltages can be written in terms of the independent twig voltages. (Fig. a)

(c) Find the ABCD parameters of the network. (Fig. b)



(a)



(b)

[7+7+7=21]

- 8) Question for extra marks. Attempt it after completing all other questions.

[5]

Suppose the output obtained in question 4 is $c(t)$. At what time (t_p) the highest value of $c(t)$ occurs? What is this highest value $c(t_p)$? Also calculate the maximum overshoot (M_p) as

$$M_p = \frac{c(t_p) - c(\infty)}{c(\infty)}$$